lemma: 6.6 [Jay Cummings]

(i) If $m \in \mathbb{Z}$, then m^2+m is even $= m(m+1)$

(ii) If $a \in \mathbb{Z}$ and a^2 is even, then a is even.

→ use contrapositive!

Proof: let $a \in \mathbb{Z}$, and a is odd.

$\therefore a = 2k+1 \quad [k \in \mathbb{Z}]$

$\therefore a^2 = (2k+1)^2 = 4k^2+4k+1 = 2\cdot(2k^2+2)+1$

$\therefore a^2$ is odd,

$\therefore \sim Q \Rightarrow \sim P$

$\therefore P \Rightarrow Q$

Proposition:

if a is an odd integer then $x^2+x-a^2=0$ has no integer solns.

$(\sim Q)$

we assume, $x^2+x-a^2=0$ has integer solns,

lets take the integer to be m,

so

$m^2+m-a^2=0$

$\Rightarrow m = \dfrac{-1 \pm \sqrt{1+4a^2}}{2\cdot 1} \quad \Rightarrow 2m+1 = \pm\sqrt{1+4a^2}$

$(2m+1)^2 = 1+4a^2$

$\Rightarrow 4m^2 + 4m + \cancel{1} = \cancel{1} + 4a^2$

$m^2 + m = a^2$

even

since, a^2 is even , a is even

a is not odd

$(\sim P)$

$(\sim Q) \Rightarrow (\sim P)$

$\therefore P \Rightarrow Q$

Modulo operation

$a \equiv b \pmod{c}$

$c \mid (a-b)$

Proposition: $a, b, c \in \mathbb{Z}$ and $n \in \mathbb{N}$,

(i) if $a \equiv b \pmod{n}$ then $a^2 \equiv b^2 \pmod{n}$

(ii) if $a \equiv b \pmod{n}$ then $ac \equiv bc \pmod{n}$

(use contrapositive) (Assignment)

Contradiction:

Someone stole Mrs Gupta's bag at 8 pm last night.

Gorman is a suspect, he was out of town yesterday at 8 pm.

So, if he were a thief, he would have been in two places at once hence this is absurd.

main idea of proof by contradiction:

Q: a statement

Q	$\sim Q$	$Q \wedge \sim Q$
T	F	F
F	T	F

$P \Rightarrow Q$ (Theorem, proposition)

P	Q	$P \Rightarrow Q$
T	F	F
F	F	T

$(P \Rightarrow Q)$	but	and
T	P : T	Q : F

Proposition:

There is no largest integer.

Proof: let's take N the largest integer.

but we know $N+1 > N$ is also an integer,

therefore N is not the largest integer,

this holds true for any $N \in \mathbb{Z}$,